Package 'asaData'

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Type Package

Title Download Datasets from 'Applied Survival Analysis'

Version 0.2.0

Description An unofficial companion package to the first and second editions of 'Applied Survival Analysis: Regression Modeling of Time to Event Data' by Hosmer and Lemeshow (1999) <ISBN:9780471154105>, and Hosmer, Lemeshow, and May (2008) <doi:10.1002/9780470258019>. Downloads and imports 16 datasets directly from Wiley companion archives, with source checksums, original numeric coding, parsed dates, and documentation of variables and edition differences. No dataset records or original codebooks are included.

License GPL-3

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Suggests survival, knitr, rmarkdown

VignetteBuilder knitr

URL https://github.com/graemeleehickey/asaData

BugReports https://github.com/graemeleehickey/asaData/issues

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`asaData-package` *Download Datasets from Applied Survival Analysis*

Description

An unofficial companion package to the first and second editions of Applied Survival Analysis: Regression Modeling of Time to Event Data.

Details

Use `asa_datasets` to list the 16 supported datasets and `download_asa` to download and import one. Each help page identifies the book edition and describes the variables, category codes, units, and endpoints. The shared `recur` and `uis` datasets use the second-edition source files. The different WHAS nested case-control samples are available as `whasncc_e1` and `whasncc_e2`.

Variable names are lower case, category codes are numeric, calendar dates are `Date` objects, and missing-value dots are represented by `NA`. Source row order and numeric values are preserved.

Source files

No dataset records or original codebooks are included in the package. Wiley archives are downloaded only when requested and cached for the current R session by default. Use `read_asa` to import an original data file already on your computer. Every import checks the source checksum, dimensions, missing-value counts, and variable types.

Licensing

Package code and original documentation are distributed under the GNU General Public License, version 3 (GPL-3). Downloaded datasets and codebooks retain their existing rights and are subject to the source terms.

Author(s)

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Source

Wiley companion data site.

UCLA companion page: analysis examples and other download formats for selected second-edition datasets.

See Also

download_asa, read_asa, asa_datasets

vignette("survival-examples", package = "asaData") for analyses using the **survival** package.

Examples

```
asa_datasets()
## Not run:
whas500 <- download_asa("whas500")
head(whas500)

## End(Not run)
```

---

actg320 *ACTG 320 Trial (Second Edition)*

Description

The 1,151-record ACTG 320 HIV treatment-trial dataset described in Table 1.5.

Format

A data frame with 1,151 rows and 16 variables:

id Integer. Participant identifier.

time Integer. Follow-up to an AIDS-defining diagnosis or death, or censoring, in days.

censor Integer. Composite endpoint indicator: 1 = AIDS-defining diagnosis or death; 0 = censored.

time_d Integer. Follow-up to death or censoring, in days.

censor_d Integer. Death endpoint indicator: 1 = death; 0 = censored.

tx Integer. Treatment indicator: 1 = regimen includes indinavir (IDV); 0 = regimen without IDV.

txgrp Integer. Treatment regimen: 1 = ZDV + 3TC; 2 = ZDV + 3TC + IDV; 3 = d4T + 3TC; 4 = d4T + 3TC + IDV. ZDV denotes zidovudine, 3TC lamivudine, and d4T stavudine.

strat2 Integer. CD4 screening stratum: 0 = CD4 count at most 50; 1 = count above 50.

sex Integer. Recorded sex: 1 = male; 2 = female.

raceth Integer. Recorded race/ethnicity: 1 = White non-Hispanic; 2 = Black non-Hispanic; 3 = Hispanic, regardless of race; 4 = Asian or Pacific Islander; 5 = American Indian or Alaskan Native; 6 = other/unknown.

ivdrug Integer. History of intravenous drug use: 1 = never; 2 = current; 3 = previous.

hemophil Integer. Haemophilia indicator: 1 = yes; 0 = no.

karnof Integer. Karnofsky Performance Scale: 100 = normal, no complaints or evidence of disease; 90 = normal activity possible with minor signs or symptoms; 80 = normal activity with effort and some signs or symptoms; 70 = self-care possible, but normal activity or active work not possible.

cd4 Numeric. Baseline CD4 count derived from multiple measurements, in cells/millilitre as stated in the codebook.

priorzdv Numeric. Duration of prior zidovudine use, in months.

age Integer. Age at enrolment, in years.

Details

Each row represents a trial participant. The composite AIDS/death endpoint is recorded in `time` and `censor`; the mortality endpoint is recorded in `time_d` and `censor_d`.

`strat2` records the screening CD4 stratum, and `cd4` records the baseline CD4 measurement.

Download

Use `download_asa("actg320")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 3, Stata examples. Table 3.2.

See Also

`actg320ncc`

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
## Not run:
actg320 <- download_asa("actg320")
library(survival)
# Second edition, Table 3.2: AIDS-defining diagnosis or death.
actg_cox <- coxph(
  Surv(time, censor == 1) ~ tx + age + sex + cd4 + priorzdv,
  data = actg320, ties = "breslow"
)
summary(actg_cox)

## End(Not run)
```

---

actg320ncc — *ACTG 320 Nested Case-Control Sample (Second Edition)*

Description

Nested case-control data for the ACTG 320 example in Section 9.4, comprising 96 matched sets.

Format

A data frame with 576 rows and 7 variables:

set Integer. Matched case-control set identifier, 1 to 96.

case Integer. Status in the sampled set: 1 = case; 0 = control.

id Integer. Original ACTG 320 participant identifier; participants may appear in more than one matched set.

time Numeric. Follow-up time, in days. The codebook states that case times were jittered to break ties.

tx Integer. Treatment indicator: 1 = regimen includes indinavir (IDV); 0 = regimen without IDV.

age Integer. Age at enrolment, in years.

cd4 Numeric. Baseline CD4 count, in cells/millilitre as stated in the codebook.

Details

Each row represents a sampled case or control. The sample contains 96 matched sets, each with one case and five controls. Participants may appear in more than one set.

`case` identifies cases and controls, and `set` identifies the matched sets. Case follow-up times in `time` are jittered to break ties.

Download

Use `download_asa("actg320ncc")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

`actg320`

Examples

```
## Not run:
actg320ncc <- download_asa("actg320ncc")
length(unique(actg320ncc$id))
table(table(actg320ncc$set))

## End(Not run)
```

asa_datasets *List the Available Applied Survival Analysis Datasets*

Description

List the datasets supported by `download_asa()` without accessing the internet.

Usage

```
asa_datasets()
```

Value

A data frame with dataset names, the source edition, dimensions, source filenames, MD5 checksums, and Wiley archive URLs. The shared `recur` and `uis` datasets use the second-edition archive.

See Also

`download_asa`, `read_asa`

Examples

```
asa_datasets()
```

bpd *Bronchopulmonary Dysplasia Data (Second Edition)*

Description

Oxygen-use follow-up for 78 subjects in the BPD dataset used in Chapter 2.

Format

A data frame with 78 rows and 4 variables:

id Integer. Subject identifier, 1 to 78.

surfact Integer. Surfactant use: 0 = no; 1 = yes.

ondays Integer. Time on oxygen, in days, to discontinuation or the end of follow-up.

censor Integer. Outcome indicator: 1 = off oxygen; 0 = still on oxygen at study end.

Details

Each row represents a subject. `ondays` records time on oxygen, and `censor == 1` indicates discontinuation of oxygen.

Download

Use `download_asa("bpd")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 2, Stata examples. Figure 2.11 and Table 2.17.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
## Not run:
bpd <- download_asa("bpd")
library(survival)
# Second edition, Figure 2.11 and Table 2.17: time to oxygen discontinuation.
bpd_km <- survfit(
  Surv(ondays, censor == 1) ~ surfact, data = bpd
)
plot(bpd_km, col = c("#0072B2", "#D55E00"), lty = 1:2,
     conf.int = FALSE, xlab = "Days on oxygen",
     ylab = "Probability of remaining on oxygen")
legend("topright", c("No surfactant", "Surfactant"),
       col = c("#0072B2", "#D55E00"), lty = 1:2, bty = "n")
bpd_logrank <- survdiff(
  Surv(ondays, censor == 1) ~ surfact, data = bpd
)
bpd_logrank

## End(Not run)
```

`comprisk` *Competing-Risks Exercise (Second Edition)*

Description

The 453-record competing-risks exercise dataset from Chapter 9.

Format

A data frame with 453 rows and 6 variables:

id Integer. Subject identifier.

age Integer. Age in years.

gender Integer. Recorded sex: 0 = male; 1 = female.

bmi Numeric. Body mass index, in kg/m^2.

time Integer. Follow-up to the recorded event or censoring, in days.

ev_typ Integer. Endpoint type: 1 = cardiovascular disease (CVD); 2 = other cause; 0 = censored.

Details

Each row represents a subject. `time` records follow-up in days. `ev_typ` distinguishes cardiovascular disease (1), other causes (2), and censoring (0).

Download

Use `download_asa("comprisk")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

`cvdrisk`

Examples

```
## Not run:
comprisk <- download_asa("comprisk")
table(comprisk$ev_typ)

## End(Not run)
```

`cvdrisk` *Cardiovascular Competing-Risks Example (Second Edition)*

Description

The 65-record CVD risk competing-risks example from Section 9.6.

Format

A data frame with 65 rows and 6 variables:

id Integer. Subject identifier.

age Integer. Age in years.

bmi Numeric. Body mass index, in kg/m^2.

gender Integer. Recorded sex: 0 = male; 1 = female.

time Numeric. Follow-up to the recorded event or censoring, in days.

ev_typ Integer. Endpoint type: 1 = cardiovascular disease (CVD); 2 = other cause; 0 = censored.

Details

Each row represents a subject. `time` records follow-up in days. `ev_typ` distinguishes cardiovascular disease (1), other causes (2), and censoring (0).

Download

Use `download_asa("cvdrisk")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

`comprisk`

Examples

```
## Not run:
cvdrisk <- download_asa("cvdrisk")
table(cvdrisk$ev_typ)

## End(Not run)
```

`download_asa` *Download an Applied Survival Analysis Dataset*

Description

Download a Wiley companion archive and return the selected dataset as an R data frame.

Usage

```
download_asa(
  name,
  cache_dir = file.path(tempdir(), "asaData"),
  refresh = FALSE,
  quiet = TRUE
)
```

Arguments

name	A single dataset name from asa_datasets(), for example "whas500".
cache_dir	Directory for downloaded ZIP archives. The default is a temporary directory for the current R session. Supply a directory of your choice to reuse downloads between sessions.
refresh	Download the archive again, even if a cached copy exists. The previous copy is retained if downloading or validating the replacement fails.
quiet	Suppress download progress messages.

Details

Wiley supplies one ZIP archive per edition. The first request for an edition downloads that archive, including its accompanying codebooks; later requests reuse it. Only the requested data file is extracted, to temporary storage. Nothing is downloaded when the package is installed or loaded.

Every returned dataset is checked against the recorded source checksum, dimensions, missing-value counts, and variable types. A changed or damaged file is rejected. Use refresh = TRUE to replace a damaged cache; if Wiley has changed the source, the package metadata will need updating.

Variable names are lower case, category codes remain numeric, calendar dates become Date objects, and source missing-value dots become NA. Row order and numeric values are preserved. The shared recur and uis datasets use the second-edition files; the two WHAS nested case-control samples have separate names.

The package contains no dataset records or original codebooks. The GPL-3 licence covers the package code and original documentation, not downloaded materials. Consult the source terms before downloading or reusing the data. A local source file can instead be imported with read_asa().

Value

A data frame. No objects are assigned to the user's workspace. Download and validation failures produce an informative error.

Source

Wiley companion data site.

See Also

asa_datasets, read_asa, whas500

Examples

```
## Not run:
whas500 <- download_asa("whas500")
head(whas500)
# The second-edition archive is reused for this request.
gbcs <- download_asa("gbcs")

## End(Not run)
```

`frtcs` *French Three Cities Study (Second Edition)*

Description

Repeated blood-pressure measurements and follow-up for 697 participants in the FRTCS dataset used in Exercise 6 of Chapter 7.

Format

A data frame with 697 rows and 17 variables:

id Integer. Participant identifier, 1 to 697.

age Integer. Age at enrolment, in years.

sex Integer. Recorded sex: 1 = male; 2 = female.

date0 Date. Baseline examination date, stored as an R Date.

sbp0 Numeric. Baseline systolic blood pressure, in mmHg.

dbp0 Numeric. Baseline diastolic blood pressure, in mmHg.

antihyp0 Integer. Antihypertensive drug-use indicator in measurement block 0: 0 = no; 1 = yes.

date1 Date. Examination date in measurement block 1, stored as an R Date.

sbp1 Numeric. Systolic blood pressure at first follow-up, in mmHg.

dbp1 Numeric. Diastolic blood pressure at first follow-up, in mmHg.

antihyp1 Integer. Antihypertensive drug use at first follow-up: 0 = no; 1 = yes.

date2 Date. Date of the second follow-up examination, stored as an R Date.

sbp2 Numeric. Systolic blood pressure at second follow-up, in mmHg. Missing values: 3.

dbp2 Numeric. Diastolic blood pressure at second follow-up, in mmHg. Missing values: 3.

antihyp2 Integer. Antihypertensive drug use at second follow-up: 0 = no; 1 = yes.

date_event Date. Date of the event or end of follow-up, stored as an R Date.

censor Integer. Outcome indicator: 1 = event; 0 = censored.

Details

Each row represents a participant, with repeated blood-pressure measurements in separate columns. Calendar follow-up in days can be calculated as `as.numeric(date_event - date0)`.

Download

Use `download_asa("frtcs")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

Examples

```
## Not run:
frtcs <- download_asa("frtcs")
with(frtcs, summary(as.numeric(date_event - date0)))

## End(Not run)
```

---

gbcs *German Breast Cancer Study (Second Edition)*

Description

Recurrence and mortality follow-up for 686 women in the German Breast Cancer Study (GBCS), described in Table 1.4.

Format

A data frame with 686 rows and 16 variables:

id Integer. Participant identifier, 1 to 686.

diagdate Date. Diagnosis date, stored as an R Date.

recdate Date. Date of recurrence or censoring, stored as an R Date; interpreted with `censrec`.

deathdate Date. Date of death or censoring, stored as an R Date; interpreted with `censdead`.

age Integer. Age at diagnosis, in years.

menopause Integer. Menopausal status: 1 = premenopausal; 2 = postmenopausal.

hormone Integer. Hormone therapy: 1 = no; 2 = yes.

size Integer. Tumour size, in millimetres.

grade Integer. Tumour grade: 1, 2, or 3, corresponding to grades I, II, and III.

nodes Integer. Number of involved lymph nodes, from 1 to 51.

prog_recp Integer. Progesterone receptor measurement.

estrg_recp Integer. Oestrogen receptor measurement.

rectime Integer. Time from diagnosis to recurrence or censoring, in days; equals `recdate - diagdate`.

censrec Integer. Recurrence indicator: 0 = censored; 1 = recurrence.

survtime Integer. Days from diagnosis to death or censoring; equals `deathdate - diagdate`.

censdead Integer. Death indicator: 0 = censored; 1 = death.

Details

Each row represents a woman. The recurrence endpoint is recorded in `rectime` and `censrec`; the mortality endpoint is recorded in `survtime` and `censdead`.

Download

Use `download_asa("gbcs")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 4, Stata examples. Table 4.11.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
## Not run:
gbcs <- download_asa("gbcs")
library(survival)
# Second edition, Table 4.11: recurrence and hormone-by-size interaction.
# Use a local 0/1 indicator: the source hormone variable is coded 1/2.
gbcs_analysis <- transform(gbcs, hormone01 = as.integer(hormone == 2))
gbcs_cox <- coxph(
  Surv(rectime, censrec == 1) ~ hormone01 * size,
  data = gbcs_analysis, ties = "breslow"
)
coef(gbcs_cox)

## End(Not run)
```

---

grace1000 — *GRACE Acute Coronary Syndrome Data (Second Edition)*

Description

The 1,000-record GRACE dataset used in Chapter 7 and described in Table 7.4.

Format

A data frame with 1,000 rows and 9 variables:

id Integer. Patient identifier, 1 to 1000.

days Numeric. Follow-up to death or censoring, in days, from 0.5 to 180. Same-day deaths or discharges are assigned 0.5 days in the source description.

death Integer. Death during follow-up: 1 = death; 0 = censored.

revasc Integer. Revascularisation performed: 1 = yes; 0 = no.

revascdays Integer. Time from admission, in days: to revascularisation if `revasc` = 1, or to recorded follow-up otherwise.

los Integer. Length of hospital stay, in days.

age Integer. Age at admission, in years.

sysbp Integer. Systolic blood pressure on admission, in mmHg.

stchange Integer. ST-segment deviation on the index electrocardiogram: 1 = yes; 0 = no.

Details

Each row represents a patient. `days` records mortality follow-up, with `death == 1` indicating death. Same-day deaths or discharges are assigned 0.5 days.

For patients with `revasc == 1`, `revascdays` gives the time from admission to revascularisation.

Download

Use `download_asa("grace1000")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 7, Stata examples. Table 7.6.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
## Not run:
grace1000 <- download_asa("grace1000")
library(survival)
# Second edition, Table 7.6: revascularisation changes during follow-up.
grace_intervals <- tmerge(
  grace1000, grace1000, id = id, death_event = event(days, death)
)
grace_intervals <- tmerge(
  grace_intervals, grace1000[grace1000$revasc == 1, ],
  id = id, revasc_t = tdc(revascdays)
)
grace_cox <- coxph(
  Surv(tstart, tstop, death_event) ~ revasc_t +
    I(1000 / age) + sqrt(sysbp) + stchange,
  data = grace_intervals, ties = "breslow"
)
c(intervals = nrow(grace_intervals), deaths = grace_cox$nevent)
summary(grace_cox)

## End(Not run)
```

`hmohiv` *Hypothetical HMO-HIV Study (First Edition)*

Description

Follow-up of 100 hypothetical HIV-positive members of a health maintenance organisation, used in Table 1.1 of the first edition.

Format

A data frame with 100 rows and 7 variables:

id Integer. Subject identifier, 1 to 100.

entdate Date. Study entry date, stored as an R Date.

enddate Date. End-of-follow-up date, stored as an R Date. For an observed death this is the event date; otherwise it is the censoring date.

time Integer. Survival or censoring time in approximate months.

age Integer. Age in years.

drug Integer. History of intravenous drug use: 0 = no; 1 = yes.

censor Integer. Follow-up outcome: 1 = death due to AIDS or AIDS-related factors; 0 = alive at study end or lost to follow-up.

Details

Each row represents a subject. `time` records survival or censoring time in approximate months, and `censor == 1` indicates death due to AIDS or AIDS-related factors. Calendar follow-up in days can be calculated as `as.numeric(enddate - entdate)`.

Download

Use `download_asa("hmohiv")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Examples

```
## Not run:
hmohiv <- download_asa("hmohiv")
with(hmohiv, table(drug, censor))
with(hmohiv, head(as.numeric(enddate - entdate)))

## End(Not run)
```

read_asa *Read a Locally Downloaded Applied Survival Analysis Dataset*

Description

Import an original Wiley '.dat' file without accessing the internet or writing a cache.

Usage

```
read_asa(name, file)
```

Arguments

name	A single dataset name from asa_datasets().
file	Path to the original, unmodified '.dat' file, extracted from the appropriate Wiley archive.

Details

The same checksum, layout, and type checks are used as for download_asa(). Files from another edition, edited files, and files with changed line endings are rejected if their checksums differ. For recur and uis, use the second-edition file.

Value

A data frame with the variables described in the corresponding dataset help page.

See Also

download_asa, asa_datasets

Examples

```
## Not run:
whas500 <- read_asa("whas500", "path/to/whas500.dat")

## End(Not run)
```

recur *Recurrent Soreness Episodes*

Description

Counting-process data from a hypothetical study of recurrent soreness episodes, used in the Chapter 9 exercises.

Format

A data frame with 1,296 rows and 7 variables:

id Integer. Subject identifier, 1 to 400; repeated across intervals.

age Integer. Age in years.

treat Integer. Treatment assignment: 0 = new treatment; 1 = old treatment.

time0 Integer. Start of the risk interval, measured in days from the study time origin; the time of the previous episode, or zero for the first interval.

time1 Integer. End of the risk interval, measured in days from the same origin; the time of the next episode or censoring.

censor Integer. Interval outcome: 1 = a soreness episode occurred at time1; 0 = censored at time1.

event Integer. Episode or risk-interval number. The supplied values are 1 to 4; the codebook gives a possible range of 0 to 4.

Details

Each row represents a risk interval for one subject. There are 1,296 intervals from 400 subjects, with up to four intervals per subject. The interval is (`time0`, `time1`], and `censor == 1` indicates an episode at its end.

The first- and second-edition files contain identical values and variable names. `recur` represents both editions using the second-edition file. The second-edition codebook specifies that `censor = 1` denotes an episode at `time1`.

Download

Use `download_asa("recur")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

Examples

```
## Not run:
recur <- download_asa("recur")
length(unique(recur$id))
head(recur[order(recur$id, recur$time0), ])

## End(Not run)
```

`uis` *UMass Impact Study of Drug Treatment*

Description

Drug-treatment follow-up data for 628 participants in the UMass Impact Study (UIS), described in Table 1.3.

Format

A data frame with 628 rows and 12 variables:

id Integer. Participant identifier, 1 to 628.

age Integer. Age at enrolment, in years. Missing values: 5.

beck Numeric. Beck Depression Score at admission, from 0 to 54. Missing values: 33.

hercoc Integer. Heroin/cocaine use during the three months before admission: 1 = both heroin and cocaine; 2 = heroin only; 3 = cocaine only; 4 = neither. Missing values: 18.

ivhx Integer. History of intravenous drug use at admission: 1 = never; 2 = previous; 3 = recent. Missing values: 18.

ndrugtx Integer. Number of previous drug treatments, from 0 to 40. Missing values: 17.

race Integer. Recorded race: 0 = White; 1 = other, using the terminology of the second-edition codebook. Missing values: 6.

treat Integer. Randomised treatment assignment: 0 = short programme; 1 = long programme.

site Integer. Treatment site: 0 = A; 1 = B.

los Integer. Length of stay in treatment, in days from admission to treatment exit.

time Integer. Time from admission to the recorded drug-use endpoint or end of follow-up, in days.

censor Integer. Event indicator: 1 = returned to drug use; 0 = otherwise, following the second-edition codebook. See the edition comparison below.

Details

Each row represents a participant. There are 628 records, of which 575 have complete data.

The two editions contain identical values in the same row and column order. `uis` represents both editions using second-edition variable names. The depression-score column is `becktota` in the first edition and `beck` in the second; this dataset uses `beck`.

The first-edition codebook defines `censor = 1` as return to drug use or loss to follow-up. The second-edition codebook describes the same indicator as return to drug use. For `race = 1`, the first edition uses the label non-White, and the second uses other.

Download

Use `download_asa("uis")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 4, Stata examples. Table 4.12.

See Also

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
## Not run:
uis <- download_asa("uis")
library(survival)
# Second edition, Table 4.12: common sample for the drug-history models.
uis_analysis <- uis[complete.cases(uis[c("age", "ivhx")]), ]
uis_analysis$drug <- as.integer(uis_analysis$ivhx %in% c(2, 3))
uis_cox <- coxph(
  Surv(time, censor == 1) ~ drug,
  data = uis_analysis, ties = "breslow"
)
c(participants = uis_cox$n, events = uis_cox$nevent)
summary(uis_cox)

## End(Not run)
```

---

whas *Worcester Heart Attack Study (First Edition)*

Description

The 481-record Worcester Heart Attack Study (WHAS) dataset described in Table 1.4 of the first edition.

Format

A data frame with 481 rows and 14 variables:

id Integer. Patient identifier, 1 to 481.

age Integer. Age recorded in the chart, in years.

sex Integer. Recorded sex: 0 = male; 1 = female.

cpk Integer. Peak cardiac enzyme measurement, in international units as stated in the source codebook.

sho Integer. Cardiogenic shock complication: 0 = no; 1 = yes.

chf Integer. Left heart failure complication: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

mitype Integer. Myocardial infarction type: 1 = Q-wave; 2 = non-Q-wave; 3 = indeterminate.

year Integer. Cohort-year code: 1 = 1975; 2 = 1978; 3 = 1981; 4 = 1984; 5 = 1986; 6 = 1988.

yrgrp Integer. Grouped cohort-year code: 1 = 1975 and 1978; 2 = 1981 and 1984; 3 = 1986 and 1988.

lenstay Integer. Length of hospital stay, in days.

dstat Integer. Status at hospital discharge: 0 = alive; 1 = dead.

lenfol Integer. Total follow-up from hospital admission to death or last follow-up, in days.

fstat Integer. Vital status at last follow-up: 0 = alive; 1 = dead.

Details

Each row represents a patient. `lenfol` and `fstat` record mortality follow-up; `lenstay` and `dstat` record the hospital-discharge endpoint.

The cohort and variable coding differ from `whas100` and `whas500`. In particular, `mitype` has three categories here and two in `whas500`.

Download

Use `download_asa("whas")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

See Also

`whas100`, `whas500`, `whasncc_e1`

Examples

```
## Not run:
whas <- download_asa("whas")
str(whas)
head(whas)

## End(Not run)
```

`whas100` *WHAS 100-Patient Dataset (Second Edition)*

Description

The 100-patient Worcester Heart Attack Study dataset described in Table 1.1.

Format

A data frame with 100 rows and 9 variables:

id Integer. Patient identifier, 1 to 100.

admitdate Date. Hospital admission date, stored as an R Date.

foldate Date. Date of death or last follow-up, stored as an R Date.

los Integer. Length of hospital stay, in days.

lenfol Integer. Follow-up from hospital admission to death or last follow-up, in days.

fstat Integer. Vital status at last follow-up: 1 = dead; 0 = alive.

age Integer. Age at admission, in years.

gender Integer. Recorded sex: 0 = male; 1 = female.

bmi Numeric. Body mass index, in kg/m^2.

Details

Each row represents a patient. `lenfol` records mortality follow-up, with `fstat == 1` indicating death.

Download

Use `download_asa("whas100")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 2, Stata examples. Table 2.3 and Figure 2.2.

See Also

`whas`, `whas500`

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
## Not run:
whas100 <- download_asa("whas100")
library(survival)
# Second edition, Table 2.3 and Figure 2.2: all-cause mortality.
whas100_km <- survfit(
  Surv(lenfol, fstat == 1) ~ 1,
  data = whas100, conf.type = "log-log"
)
summary(whas100_km, times = c(6, 14, 44))
```

```
plot(whas100_km, xscale = 365.25, conf.int = FALSE,
     xlab = "Years since admission", ylab = "Survival probability")

## End(Not run)
```

`whas500` *WHAS 500-Patient Dataset (Second Edition)*

Description

The 500-patient Worcester Heart Attack Study dataset described in Table 1.2.

Format

A data frame with 500 rows and 22 variables:

id Integer. Patient identifier, 1 to 500.

age Integer. Age at hospital admission, in years.

gender Integer. Recorded sex: 0 = male; 1 = female.

hr Integer. Initial heart rate, in beats per minute.

sysbp Integer. Initial systolic blood pressure, in mmHg.

diasbp Integer. Initial diastolic blood pressure, in mmHg.

bmi Numeric. Body mass index, in kg/m^2.

cvd Integer. History of cardiovascular disease: 0 = no; 1 = yes.

afb Integer. Atrial fibrillation: 0 = no; 1 = yes.

sho Integer. Cardiogenic shock: 0 = no; 1 = yes.

chf Integer. Congestive heart failure complication: 0 = no; 1 = yes.

av3 Integer. Complete heart block: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

mitype Integer. Myocardial infarction type: 0 = non-Q-wave; 1 = Q-wave.

year Integer. Cohort-year code: 1 = 1997; 2 = 1999; 3 = 2001.

admitdate Date. Hospital admission date, stored as an R Date.

disdate Date. Hospital discharge date, stored as an R Date.

fdate Date. Date of death or last follow-up, stored as an R Date.

los Integer. Length of hospital stay, in days, from admission to discharge.

dstat Integer. Hospital discharge status: 0 = alive; 1 = dead.

lenfol Integer. Total follow-up from admission to death or last follow-up, in days.

fstat Integer. Vital status at last follow-up: 0 = alive; 1 = dead.

Details

Each row represents a patient. `lenfol` and `fstat` record mortality follow-up; `los` and `dstat` record the hospital-discharge endpoint.

The coding of `mitype` and `year` differs from that in `whas`.

Download

Use `download_asa("whas500")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

UCLA companion page: SAS and SPSS downloads and analysis examples for Stata, SPSS, and SAS.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

UCLA Office of Advanced Research Computing. Applied Survival Analysis, second edition: Chapter 5, Stata examples. Table 5.3.

See Also

`whas`, `whas100`, `whasncc_e2`

`vignette("survival-examples", package = "asaData")` for longer analyses.

Examples

```
## Not run:
whas500 <- download_asa("whas500")
library(survival)
# Second edition, Table 5.3: all-cause mortality after admission.
whas500_cox <- coxph(
  Surv(lenfol, fstat == 1) ~ age + hr + sysbp + diasbp + bmi +
    gender + cvd + afb + chf + miord + mitype,
  data = whas500, ties = "breslow"
)
summary(whas500_cox)

## End(Not run)
```

---

whasncc_e1 *WHAS Nested Case-Control Sample (First Edition)*

Description

Nested case-control data for the Worcester Heart Attack Study, used in the Chapter 9 exercises.

Format

A data frame with 1,494 rows and 12 variables:

set Integer. Matched case-control set identifier.

case Integer. Status within this sampled set: 0 = control; 1 = case.

t Numeric. Follow-up time with ties broken, in days.

lenfol Integer. Total follow-up from hospital admission, in days.

fstat Integer. Vital status at last follow-up in the cohort: 0 = alive; 1 = dead.

age Integer. Age recorded in the chart, in years.

sex Integer. Recorded sex: 0 = male; 1 = female.

chf Integer. Left heart failure complication: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

yrgrp Integer. Grouped cohort-year code: 1 = 1975 and 1978; 2 = 1981 and 1984; 3 = 1986 and 1988.

lenstay Integer. Length of hospital stay, in days.

nr Integer. Size of the underlying cohort risk set at sampling. Missing values: 858.

Details

Each row represents a sampled case or control within a risk set. There are 249 matched sets, each with one case and five controls. Individuals may appear in multiple risk sets.

`case` identifies cases and controls, and `set` identifies the matched sets. `fstat` records vital status in the underlying cohort.

The editions contain different sampled records and variables. The first edition has 1,494 rows in 249 matched sets and 12 variables; the second has 1,638 rows in 273 sets and 11 variables. The second edition includes `bmi` but omits `yrgrp` and `lenstay`.

Download

Use `download_asa("whasncc_e1")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr. and Lemeshow, S. (1999). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. First edition. Wiley. ISBN 978-0-471-15410-5.

See Also

`whasncc_e2`

Examples

```
## Not run:
whasncc_e1 <- download_asa("whasncc_e1")
table(table(whasncc_e1$set))
with(whasncc_e1, table(case, fstat))

## End(Not run)
```

whasncc_e2 *WHAS Nested Case-Control Sample (Second Edition)*

Description

Nested case-control data for the Worcester Heart Attack Study, used in the Chapter 9 exercises.

Format

A data frame with 1,638 rows and 11 variables:

set Integer. Matched case-control set identifier.

case Integer. Status within this sampled set: 0 = control; 1 = case.

t Numeric. Follow-up time with ties broken, in days.

lenfol Integer. Total follow-up from hospital admission, in days.

fstat Integer. Vital status at last follow-up in the cohort: 0 = alive; 1 = dead.

age Integer. Age recorded in the chart, in years.

sex Integer. Recorded sex: 0 = male; 1 = female.

bmi Numeric. Body mass index, in kg/m^2.

chf Integer. Left heart failure complication: 0 = no; 1 = yes.

miord Integer. Myocardial infarction order: 0 = first; 1 = recurrent.

nr Integer. Size of the underlying cohort risk set at sampling. Missing values: 1365.

Details

Each row represents a sampled case or control within a risk set. There are 273 matched sets, each with one case and five controls. Individuals may appear in multiple risk sets.

`case` identifies cases and controls, and `set` identifies the matched sets. `fstat` records vital status in the underlying cohort.

The editions contain different sampled records and variables. The first edition has 1,494 rows in 249 matched sets and 12 variables; the second has 1,638 rows in 273 sets and 11 variables. The second edition includes `bmi` but omits `yrgrp` and `lenstay`.

Download

Use `download_asa("whasncc_e2")` to retrieve this dataset. Data are not included in the package.

Source

Wiley companion data site.

References

Hosmer, D. W., Jr., Lemeshow, S., and May, S. (2008). *Applied Survival Analysis: Regression Modeling of Time to Event Data*. Second edition. Wiley. doi:10.1002/9780470258019.

See Also

whasncc_e1

Examples

```
## Not run:
whasncc_e2 <- download_asa("whasncc_e2")
table(table(whasncc_e2$set))
with(whasncc_e2, table(case, fstat))

## End(Not run)
```

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